

Lecture 09-09: Linear, 2-D

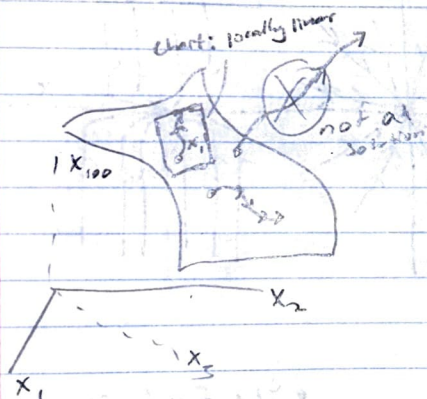
5. Linear, 2-D systems

$$\begin{aligned} \dot{x} &= ax + by \\ \dot{y} &= cx + dy \end{aligned} \rightarrow (x, y) \equiv \vec{x}: \quad \dot{\vec{x}} = \underline{A} \cdot \vec{x}; \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

convenient moving forward to use matrix representation.

linear systems: $\{x_1(t), x_2(t)\}$ are solutions, this implies

$\Rightarrow Kx_1(t) + Lx_2(t)$ is also a solution



$\rightarrow$ higher dimensional systems rely heavily on linear approximations

$\rightarrow$ visualizing > 3 dims is possible but tricky

$\rightarrow$ linearity in 2D manifolds is facilitated by a covering with locally linear charts

Section 5.1.1: Harmonic Oscillator

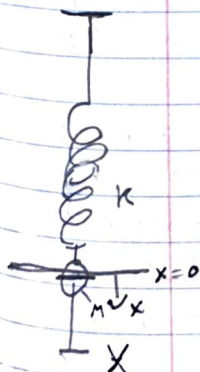
$$m\ddot{x} + kx = 0$$

$\rightarrow$ State space $\rightarrow$

$$\dot{\vec{x}} = \vec{v}$$

$$\dot{v} = -\frac{k}{m}x = -\omega^2 x$$

phase plane / phase space



The second dimension allows for periodic orbits!

- $\rightarrow$ trajectories in phase space are ellipses! (due to Conserv. of E)
- $\rightarrow$ trajectories always rotate clockwise in phase space
- $\rightarrow$ for every initial condition, there is an elliptical / periodic orbit that satisfies the equation
- $\rightarrow \omega^2 x^2 + v^2 = \text{const.}$



"The world decided that it is always foliated into energy shells."

Section 5.1.2

$$\dot{\vec{x}} = \underline{A} \cdot \vec{x}$$

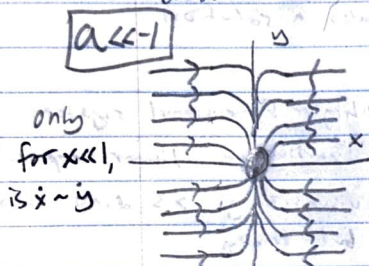
consider the case of an uncoupled system:

$$\begin{aligned} \dot{x} &= ax \\ \dot{y} &= -y \end{aligned}$$

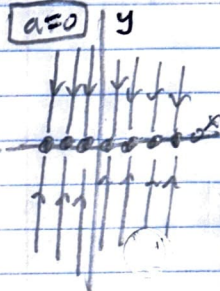
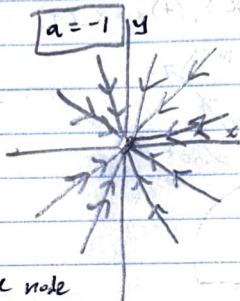
$$\underline{A} = \begin{pmatrix} a & 0 \\ 0 & -1 \end{pmatrix}$$

$$\Rightarrow x(t) = e^{at} x_0, y(t) = e^{-t} y_0 \Rightarrow \vec{x}(t) = \vec{x}_0 e^{\begin{pmatrix} a \\ -1 \end{pmatrix} t}$$

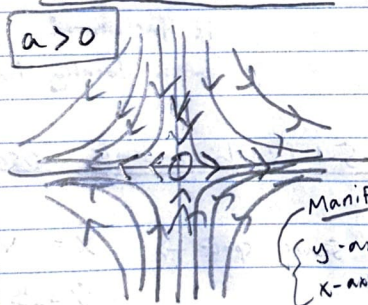
consider the 4 cases of the system, with different $a \in \mathbb{R}$



$x^* = 0$ is a stable node



line of fixed points! $x = x_0$ ($\dot{x} = 0$)



Manifolds

$\begin{cases} y\text{-axis is stable} \\ x\text{-axis is unstable} \end{cases}$

Saddle-node
(bifurcation)

• the fixed point $x^* = 0$ is stable (node/manifold) for all cases but when $a > 0$

• $a = -1 \Rightarrow x^*$ is a star node or symmetrical node

• In 2D systems, stable nodes can bifurcate into Manifolds

Now, we consider a system where $\underline{A}$ includes the coupling elements (off-diagonal)

Section 5.2: Classification

$$\underline{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{let } \underline{\dot{x}} = \underline{A} \cdot \underline{x}$$

Matrix properties $\Rightarrow \underline{A} \underline{v} = \lambda \underline{v} \Rightarrow (\underline{A} - \lambda \underline{I}) \underline{v} = 0$
imply λ eigenvalue $\underline{v}$ eigenvectors

$$\Rightarrow \det(\underline{A} - \lambda \underline{I}) = 0 \text{ is the characteristic equation}$$

$$\Rightarrow \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0 \Rightarrow (a-\lambda)^2 - cb = 0$$

$$\Rightarrow \lambda^2 - \text{Tr} \underline{A} \lambda + \det \underline{A} = 0$$